Mohandi Anis

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PROFESSIONAL SUMMARY

• Master's student in Intelligent Embedded Systems and Quantum Informatics with a background in telecommunications engineering and machine learning.

• Research experience in reinforcement learning, signal processing, and deep learning, with applications in communication systems and forecasting.

• Developed ML-based solutions including adaptive quantization for FBMC systems using Deep Q-Learning and hybrid time-series forecasting models.

• Strong programming background in Python, PyTorch, TensorFlow, and C++, with experience implementing and evaluating machine learning algorithms.

• Motivated to pursue research at the intersection of machine learning and communication systems, with the goal of developing efficient intelligent signal processing methods.

EDUCATION

Budapest University of Technology and Economics — Budapest, HU

M.Sc. in Intelligent Embedded Systems and Quantum Informatics — Jan 2026

Thesis: Adaptive Quantizer for FBMC Prototype Filters Using Reinforcement Learning

GPA: 4.16

University of Bechar — Bechar, DZ

B.Sc. in Telecommunications Engineering — Jun 2023

GPA: 3.7

TECHNICAL RESEARCH EXPERIENCE

Adaptive Quantizer for FBMC Prototype Filters Using Reinforcement Learning

Master Thesis Project

• Developed a Deep Q-Learning agent to optimize quantization parameters in Filter Bank Multicarrier communication systems.

• Implemented reinforcement learning techniques including experience replay and epsilon-greedy exploration.

• Achieved 12% reduction in operation time compared to traditional quantization approaches.

• Investigated trade-offs between quantization precision and communication system performance.

Quantization Effects in FBMC Communication Systems

• Conducted experimental study on finite-precision effects in FBMC communication systems.

• Compared uniform and non-uniform quantization techniques.

• Analyzed impact on spectral efficiency, signal reconstruction, and system robustness.

ACADEMICAL PROJECTS

Hybrid Statistical–Deep Learning Model for Long-Term Electricity Demand Forecasting

• Developed a hybrid forecasting framework coupling a statistical time-series model to capture seasonal and trend components with a Gated Recurrent Unit neural network to model nonlinear residual patterns.

• Designed a double stage learning pipeline where the GRU model learns the residual error of the statistical model and corrects its predictions, which leads to improving long term forecasting.

• Evaluated the model for 3-month ahead electricity demand forecasting, it demonstrated effectiveness of hybrid statistical deep learning approaches for long-term energy demand prediction.

Deep Learning-Based Chess Piece Segmentation Using Semantic Image Segmentation

• Developed a semantic segmentation framework for pixel-level identification and classification of chess pieces from chessboard images.

• Implemented a DeepLabV3 convolutional neural network architecture to capture fine-grained object boundaries and complex piece geometries.

• Utilized a publicly available chess dataset and applied data augmentation techniques as in rotation, illumination variation, and perspective changes to improve model generalization under furthermore varying inputs.

• Designed and trained the system to produce pixel-wise segmentation masks, enabling accurate localization and classification of individual chess pieces.

TECHNICAL SKILLS & AWARDS

Computer Skills

Programming Languages

Python, C++, SQL

Machine Learning & AI

PyTorch, TensorFlow, Reinforcement Learning, Deep Learning, LLM, RAG pipelines.

Signal Processing & Communications

Digital Signal Processing, FBMC Communication Systems

Tools & Platforms

Git, AWS, .NET Framework

Other Technical Skills

Web Scraping, Data Processing, Model Training & Evaluation

Languages

English: Advanced (C1)
French: Native
Hungarian: Basic (A2)
Arabic: Native

Honors & Awards

• **STIPENDIUM HUNGARICUM Scholarship Awardee 2023 (Masters)**
• **Valedictorian of the 2020 Bachelor of Telecommunications Class (Bachelor)**